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Java SE 27 & JDK 27

Class InheritableThreadLocal<T>

java.lang.Object
    java.lang.ThreadLocal<T>
        java.lang.InheritableThreadLocal<T>

Type Parameters:

T - the type of the inheritable thread local's value

```
public class InheritableThreadLocal<T>
extends ThreadLocal<T>
```

This class extends `ThreadLocal` to provide inheritance of values from parent thread to child thread: when a child thread is created, the child receives initial values for all inheritable thread-local variables for which the parent has values. Normally the child's values will be identical to the parent's; however, the child's value can be made an arbitrary function of the parent's by overriding the `childValue` method in this class.

Inheritable thread-local variables are used in preference to ordinary thread-local variables when the per-thread-attribute being maintained in the variable (e.g., User ID, Transaction ID) must be automatically transmitted to any child threads that are created.

Note: During the creation of a new `thread`, it is possible to *opt out* of receiving initial values for inheritable thread-local variables.

Since:

1.2

See Also:

`ThreadLocal`,
`Thread.Builder.inheritInheritableThreadLocals(boolean)`

Constructor Summary

Constructors

Constructor	Description
`InheritableThreadLocal()`	Creates an inheritable thread local variable.

Method Summary

All Methods | **Instance Methods**

Modifier and Type	Method	Description

protected T	**childValue**(T parentValue)	Computes the child's initial value for this inheritable thread-local variable as a function of the parent's value at the time the child thread is created.

Methods declared in class ThreadLocal

`get`, `initialValue`, `remove`, `set`, `withInitial`

Methods declared in class Object

`clone`, `equals`, `finalize`, `getClass`, `hashCode`, `notify`, `notifyAll`, `toString`, `wait`, `wait`, `wait`

Constructor Details

InheritableThreadLocal

```
public InheritableThreadLocal()
```

Creates an inheritable thread local variable.

Method Details

childValue

```
protected T childValue(T parentValue)
```

Computes the child's initial value for this inheritable thread-local variable as a function of the parent's value at the time the child thread is created. This method is called from within the parent thread before the child is started.

This method merely returns its input argument, and should be overridden if a different behavior is desired.

Parameters:

`parentValue` - the parent thread's value

Returns:

the child thread's initial value

Report a bug or suggest an enhancement

For further API reference and developer documentation see the Java SE Documentation, which contains more detailed, developer-targeted descriptions with conceptual overviews, definitions of terms, workarounds, and working code examples. Other versions.

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